Compiler Design

CHAPTER 7

Runtime Environment & Code Optimization

Neso Academy

Compiler Design

Runtime Environment

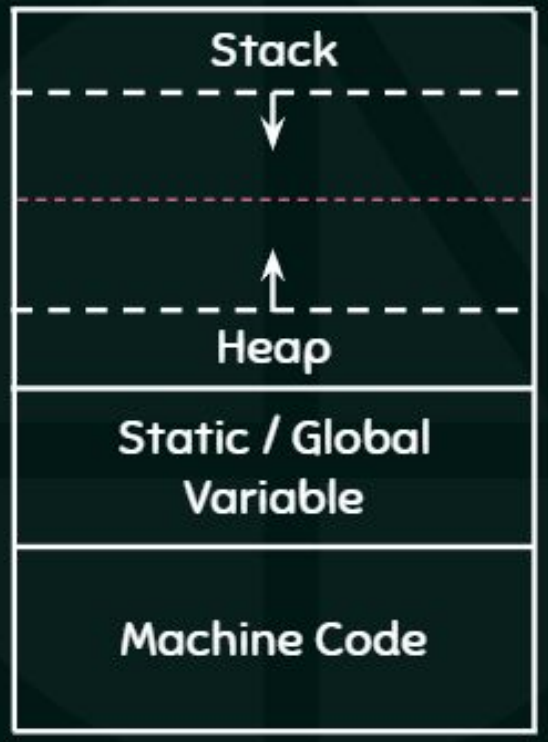
Outcome

- ☆ Different Storage Allocation Strategies.

OPERATING SYSTEM
PRINTER

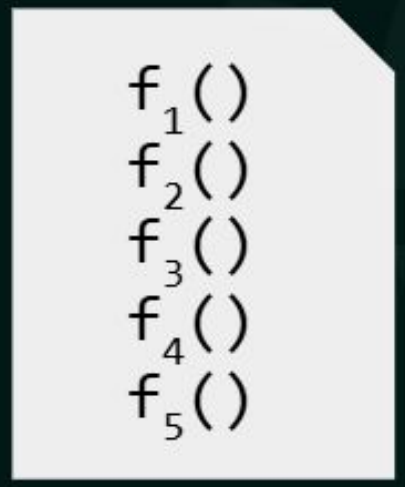
</CODE>

Memory Layout:

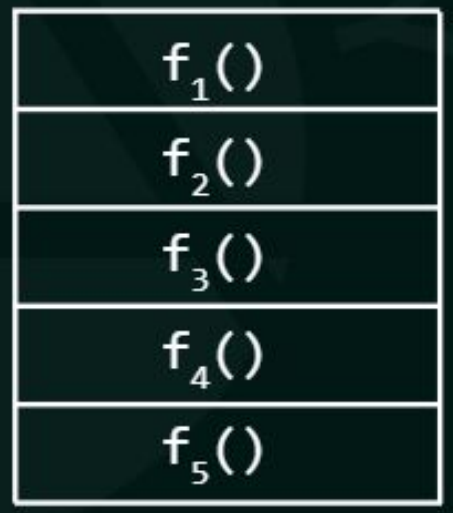


Storage Allocation Strategies:

1. Static:
   i. Allocation is done at Compile time.
   ii. Bindings do not change at Run time.
   iii. One activation record per procedure.



Program



Static Allocation

Storage Allocation Strategies:

1. Static:
   i. Allocation is done at Compile time.
   ii. Bindings do not change at Run time.
   iii. One activation record per procedure.

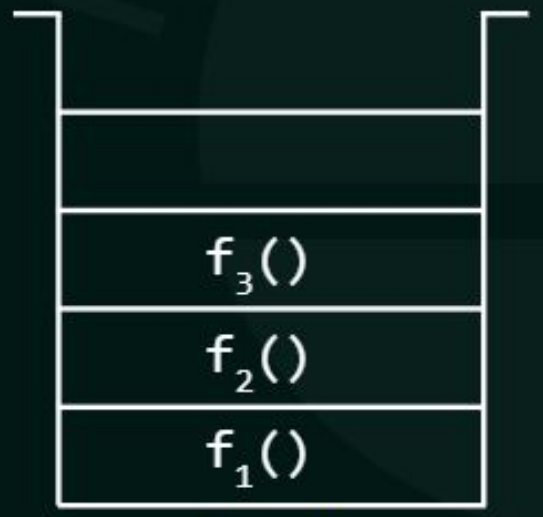
- Cons:
  i. Recursion is not supported.
  ii. Size of data objects must be known at Compile time.
  iii. Data Structures cannot be created dynamically.

- Pro: The element which is provided with the static allocation gets the lifetime as same as the process itself.

📌 *Note:* Global Arrays are static by default.

Storage Allocation Strategies:

2. Stack: Whenever a new activation begins, the activation record is pushed onto the stack and whenever activation ends, the activation record is popped off.



Stack

```
f1()
{
    f2()
    {
        f3()
        {

        }
    }
}
```

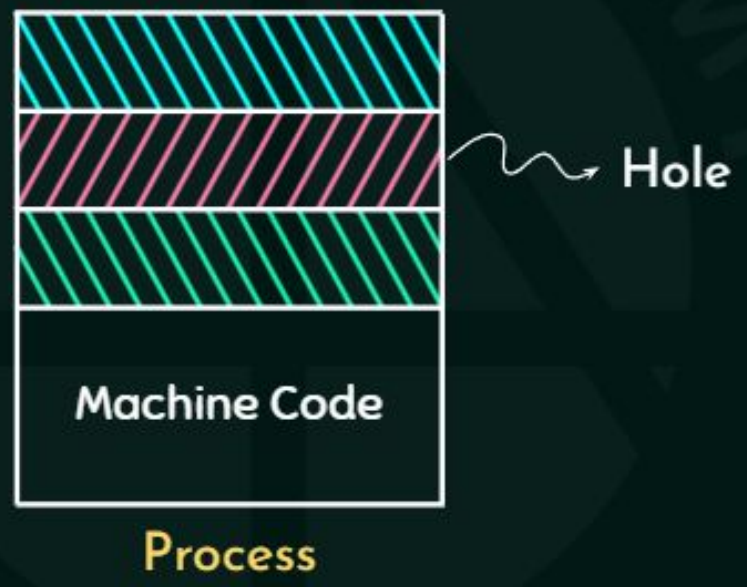
Storage Allocation Strategies:

2. Stack: Whenever a new activation begins, the activation record is pushed onto the stack and whenever activation ends, the activation record is popped off.

- Con: Local variables cannot be retrieved once activation ends.
- Pro: Recursion is supported.

Storage Allocation Strategies:

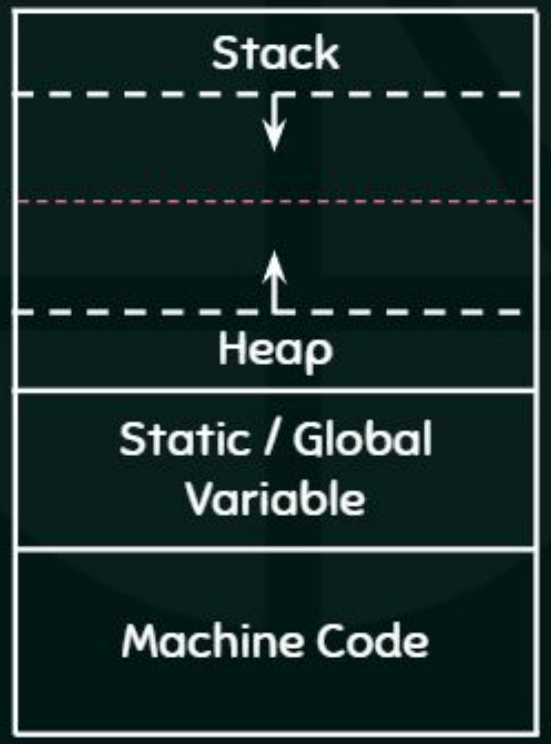
3. **Heap:** Allocation and deallocation can be done any order.
   - **Con:** Heap management is an overhead.



📌 *Note:* In C programming, First Fit is used.

Storage Allocation Strategies – Summary:

1. Permanent lifetime in case of static allocation.
2. Nested lifetime in case of stack allocation.
3. Arbitrary lifetime in case of heap allocation.



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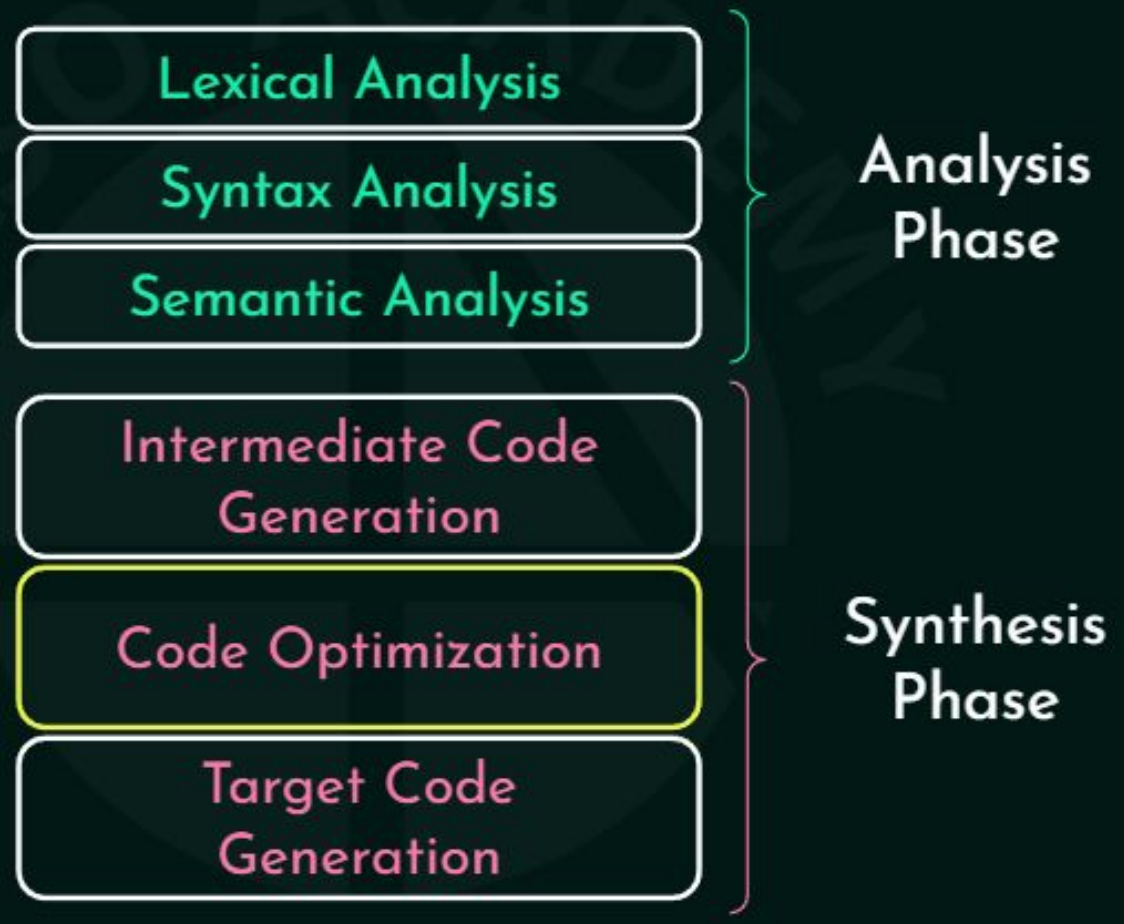
Code Optimization

Outcome

- ☆ Understanding the purpose of Code Optimization.
- ☆ Objective of Code Optimization.
- ☆ Different Code Optimization techniques.

Compiler – Internal Architecture



Objective of Code Optimization:

1. The optimization must be correct and must not change the meaning of the program.
2. The compilation time must be kept reasonable.
3. The optimization process should not delay the overall compiling process.
4. Optimization should increase the speed and performance of the program.

Purpose of Code Optimization:

1. It is used to reduce the consumed memory space.
2. It is used to increase the compilation speed.
3. An optimized code facilitates re-usability.

Types of Code Optimization:

- Machine Independent:
  - Improves the intermediate code.
- Machine Dependent:
  - It involves CPU registers and may have absolute memory references rather than relative references.
  - It is performed after the target code has been generated.

Types of Code Optimization:

Machine Independent Optimizations

1. Loop Optimizations

Machine Dependent Optimizations

Loop Optimization:

```
for(int i=0;i<10;i++)
{
    a = i*2;
}

for(int j=0;j<10;j++)
{
    b = j+3;
}
```

```
for(int i=0;i<10;i++)
{
    a = i*2;
    b = i+3;
}
```

Types of Code Optimization:

Machine Independent Optimizations

1. Loop Optimizations
2. Constant Folding
3. Constant Propagation
4. Operator Strength Reduction
5. Dead Code Elimination
6. Common Subexpression Elimination
7. Algebraic Simplification

Machine Dependent Optimizations

1. Register Allocation
2. Instruction Scheduling
3. Peephole Optimizations
   a. Redundant LOAD and STORE
   b. Flow Control Optimizations
   c. Use of Machine Idioms

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Loop Optimization – Part 1

Outcome

- ☆ Requisites for Loop Optimization.
- ☆ Understanding Basic blocks, Program Flow Graph and Control Flow Analysis.

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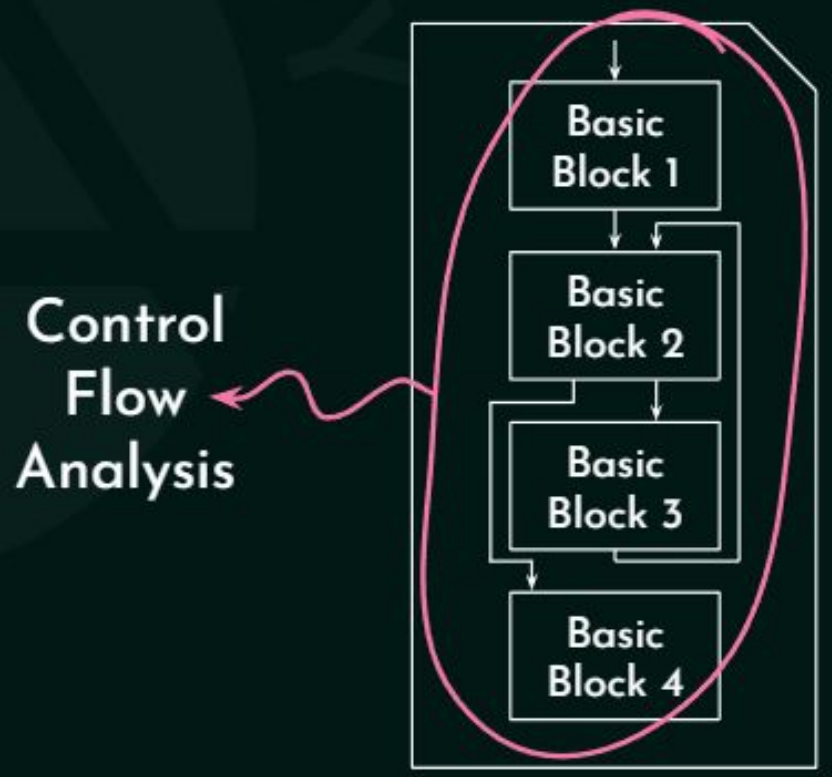
Loop Optimization:

- The loops must be detected.
- Loops are detected through Control Flow Analysis(CFA) using Program Flow Graphs(PFG).
- In order determine PFG, we first need to detect the Basic Blocks.

A Basic Block is a sequence of 3-address statements where control enters at the beginning and leaves only from the end without any jumps or halts.



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Loop Optimization – Part 2

Outcome

- ☆ How to determine the Basic Blocks.
- ☆ Illustration of Control Flow Analysis.

Loop Optimization:

- The loops must be detected.
- Loops are detected using Control Flow Analysis(CFA) using Program Flow Graphs(PFG).
- In order determine PFG, we first need to detect the Basic Blocks.

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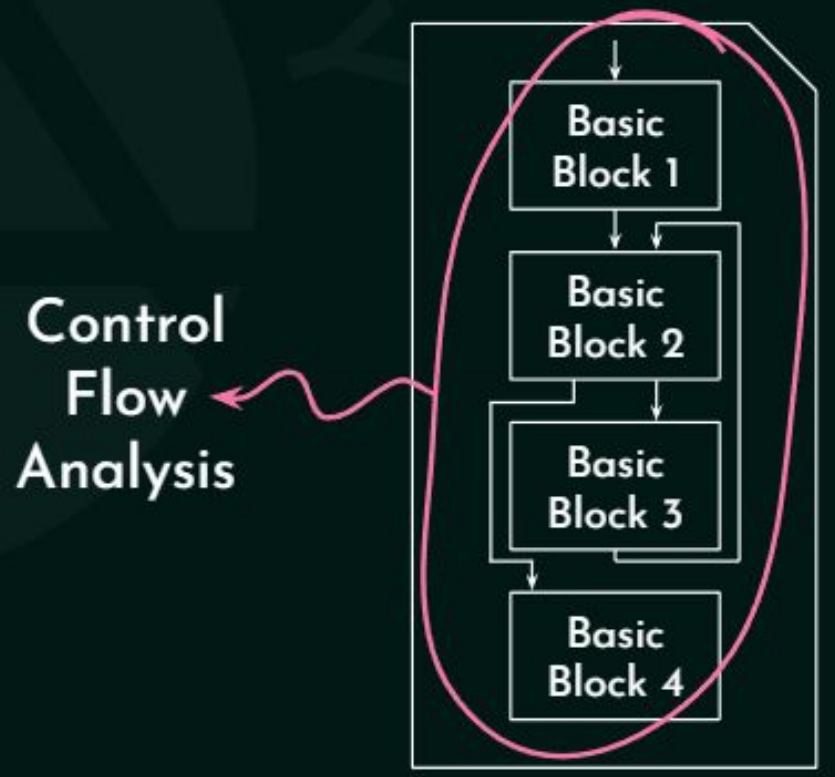

How to find the Basic Blocks?

-- Find the Leader.

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-- Find the Leader.

Identifying the Leader:

1. First statement is a leader.
2. Statement that is the target of a conditional or unconditional GOTO is a leader.
3. Statement that immediately follows a conditional or unconditional GOTO is a leader.

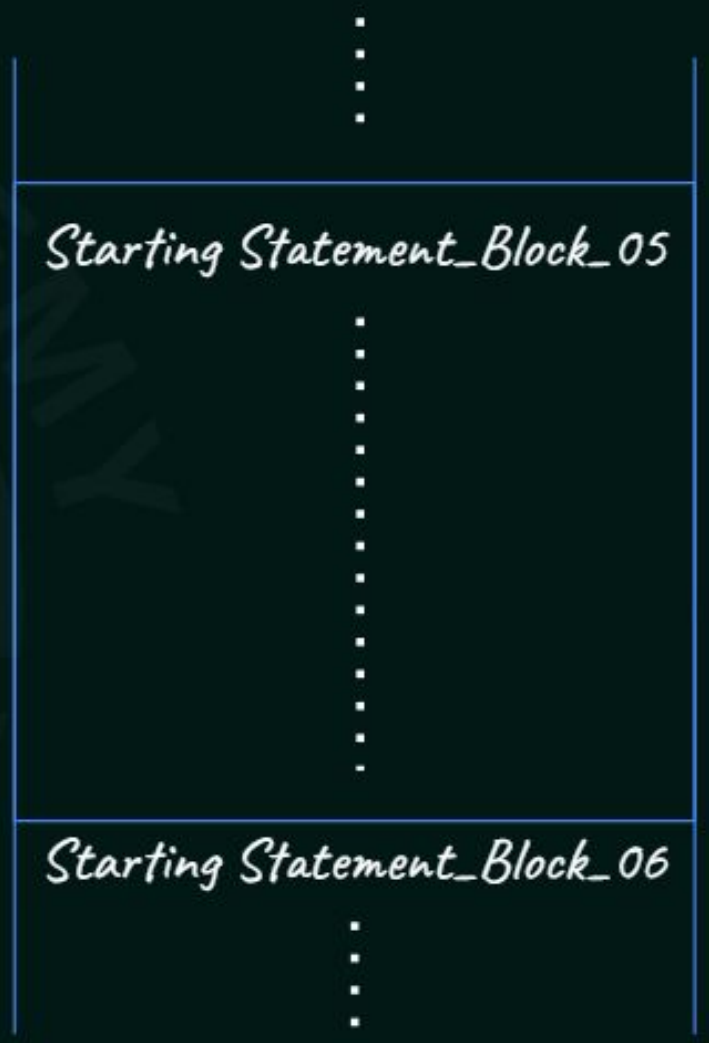

Illustration – Producing PFG:

HLL code:

```
fact(a){
    int f = 1;
    for(i=2;i<=a;i++)
        f = f * i;
}
```

TAC:

1. $f = 1$
2. $i = 2$
3. if(i<=a) GOTO 9
4. $t_1 = f * i$
5. $f = t_1$
6. $t_2 = i + 1$
7. $i = t_2$
8. GOTO 3
9. GOTO <Calling Program>

Illustration – Producing PFG:

Identifying the Leader:

1. First statement is a leader.
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TAC:

```
1.  f = 1                    BB1
2.  i = 2
3.  if(i<=a) GOTO 9          BB2
4.  t1 = f * i               BB3
5.  f = t1
6.  t2 = i + 1
7.  i = t2
8.  GOTO 3
9.  GOTO <Calling Program>   BB4
```

Illustration – Producing PFG:

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```

PFG:

Illustration – Performing CFA:

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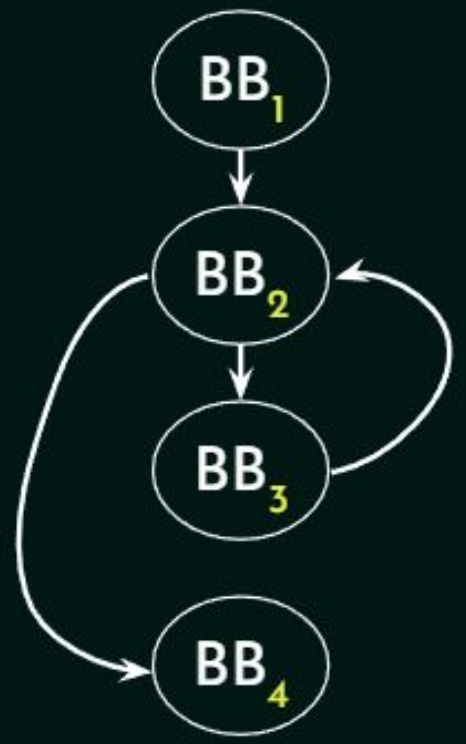
PFG:



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Loop Optimization Techniques

Outcome

- ☆ Different types of Loop Optimization Techniques.

Loop Optimization Technique – Code Motion:

- The number of statements within the loop is reduced.
- Also known as Frequency reduction.

```
while(i<1000)
{
    a = (sin(x)/cos(x)) * i;
    i++;
}
```

→

```
t = sin(x)/cos(x);
while(i<1000)
{
    a = t * i;
    i++;
}
```

Note: Moving the expression with computation outside, is a.k.a. Loop Invariant Method.

Loop Optimization Technique – Loop Unrolling:

- If possible, eliminate the entire loop.

```
for(int i=0;i<5;i++)
{
    printf("Hello");
}
```

```
printf("Hello");
printf("Hello");
printf("Hello");
printf("Hello");
printf("Hello");
```

Loop Optimization Technique – Loop Jamming:

- Combine the loop bodies.
- Also known as Loop Fusion.

```
for(int i=0;i<10;i++)
{
    a = i*2;
}

for(int j=0;j<10;j++)
{
    b = j+3;
}
```

```
for(int i=0;i<10;i++)
{
    a = i*2;
    b = i+3;
}
```

Note: The opposite transformation is called Loop fission or Loop distribution.

Loop Optimization Technique – Loop Unswitching:

- It lifts conditions out of loops, creating two loops.

```
for(i=0;i<100;++i)
{
    if(c)
    {
        f();
    }
    else
    {
        g();
    }
}
```

```
if(c)
{
    for(i=0;i<100;++i)
        f();
}
else
{
    for(i=0;i<100;++i)
        g();
}
```

Note: Also known as Loop splitting.

Loop Optimization Technique – Loop Peeling:

- Here problematic iteration is resolved separately before entering the loop.

```
for(i=0;i<10;i++)
{
    if(i==0)
    a[i] = . . .
    else
    b[i] = . . .
}
```

→

```
a[0] = . . .
for(i=1;i<10;i++)
{
    b[i] = . . .
}
```

Compiler Design

Machine Independent Optimization Techniques

Outcome

- ☆ Different Machine Independent Optimization Techniques.

Types of Code Optimization:

Machine Independent Optimizations

1. Loop Optimizations
2. Constant Folding
3. Constant Propagation
4. Operator Strength Reduction
5. Dead Code Elimination
6. Common Subexpression Elimination
7. Algebraic Simplification

Machine Dependent Optimizations

1. Register Allocation
2. Instruction Scheduling
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Constant Folding:

- Evaluation of expressions at compile-time.
- Applicable to the operands which are known to be constant.

```
a = 10 * 5 + 6 - b;
```

```
a = 56 - b;
```

Constant Propagation:

- If a variable is assigned a constant value, then subsequent uses of that variable can be replaced by the constant as long as no intervening assignment has changed the value of the variable.

1.
```
a = 13.7
b = a/4.5
```
→ `a/4.5 as 13.7/4.5`

2.
```
j = 1
if(j) GOTO L
```
→ `GOTO L`

Operator Strength Reduction:

- It replaces an operator by a less expensive one.

```
b = a * 2   ➡   b = a << 1
```

Dead Code Elimination:

- If an instruction's result is never used, the instruction is considered dead and can be removed from the instruction stream.

$$\cancel{t_1 = t_2 \times t_3}$$

- If t_1 holds the result of a function call, we cannot eliminate the instruction.

Common Subexpression Elimination:

- Two operations are common if they produce the same result. In such a case, it is likely more efficient to compute the result once and reference it the second time rather than re-evaluate it.

```
A = B + C
D = 2 + B + 3 + C
```

→

```
A = B + C
D = 2 + 3 + A
```

Algebraic Simplification:

- Simplifications use algebraic properties or particular operator-operand combinations to simplify expressions.
- These optimizations can remove useless instructions entirely via algebraic identities.

```
A = A + 0
B = B * 1
```

Compiler Design

Machine Dependent Optimization Techniques

Outcome

- ☆ Different Machine dependent Optimization Techniques.

Types of Code Optimization:

Machine Independent Optimizations

1. Loop Optimizations ✓
2. Constant Folding ✓
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Register Allocation:

- The most effective optimization for all architectures.
- Solely depends on the number of available registers.
- Types:
    a. Local Allocation
    b. Global Allocation

Instruction Scheduling:

- Using this the pipelining capability of the architecture can be used effectively.
- Instructions can be placed in the delay slots (like NOOP).

Peephole Optimization – Redundant LOAD & STORE:

```
x = y + z

MOV y, R0
ADD z, R0
MOV R0, x
```

```
a = b + c
d = a + e
```

```
MOV b, R0
ADD c, R0
MOV R0, a

MOV a, R0
ADD e, R0
MOV R0, d
```

Redundant !!

Peephole Optimization – Flow Control:

- Avoid jumps on jumps:

```
L1: GOTO L2
      ⋮
L2: GOTO L3
      ⋮
L3: GOTO L4
      ⋮
L4:
```

→

```
L1: GOTO L4
      ⋮
L4:
```

- Eliminate Dead Code:

```
#define x 0
if(x)
{
    ...
}
```

Peephole Optimization – Use of Machine Idioms:

```
i = i + 1      MOV i, R0
               ADD 1, R0        INC i
               MOV R0, i
```

Compiler Design

Liveness Analysis

Outcome

- ☆ Understanding Liveness Analysis.
- ☆ Solved problem on Liveness Analysis.

Objective of Code Optimization:

1. The optimization must be correct and must not change the meaning of the program.
2. The compilation time must be kept reasonable.
3. The optimization process should not delay the overall compiling process.
4. Optimization should increase the speed and performance of the program.

Liveness Analysis:

- What is Liveness?

  A variable is live if its value will be used before it gets overwritten.

- Why is it important?

Types of Code Optimization:

Machine Independent Optimizations

1. Loop Optimizations
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Liveness Analysis:

What is Liveness?

A variable is live if its value will be used before it gets overwritten.

Why is it important?

- Register allocation: It helps in deciding which variables to keep in registers and which to store in memory to optimize performance.
- Dead code elimination: It can be used to identify and remove code that computes values that are never used.
- Code scheduling: It's used to reorder instructions to minimize stalls and improve instruction-level parallelism.

Illustration – Performing CFA:

HLL code:

```
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    int f = 1;
    for(i=2;i<=a;i++)
        f = f * i;
}
```

TAC:

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1.  f = 1                    BB1
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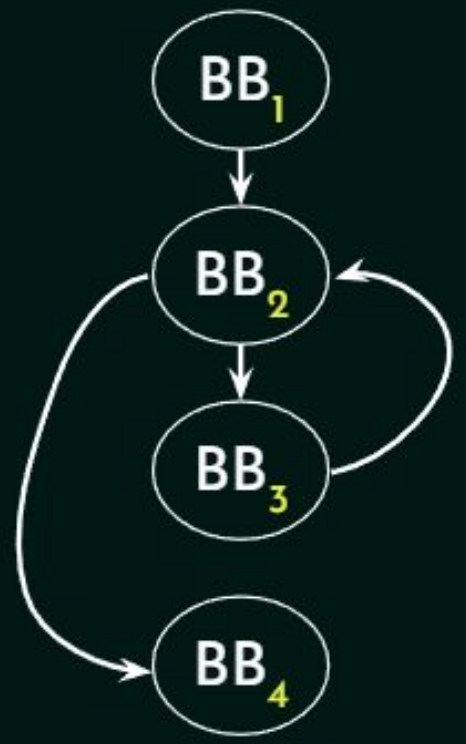
PFG:



Liveness Analysis – Algorithm:

- **Input:** Program Flow Graph
- **Output:** Liveness Information
  - -- IN[B] (Set of variables that are live at the beginning of the Block)
  - -- OUT[B] (Set of variables that are live after the Block)
  - -- DEF/KILL[B] (Set of variables that are defined/killed in the Block)
  - -- USE/GEN[B] (Set of variables that are used/generated in the Block)

Liveness Analysis – Algorithm:

1. **Initialization:** IN and OUT sets are initially empty.

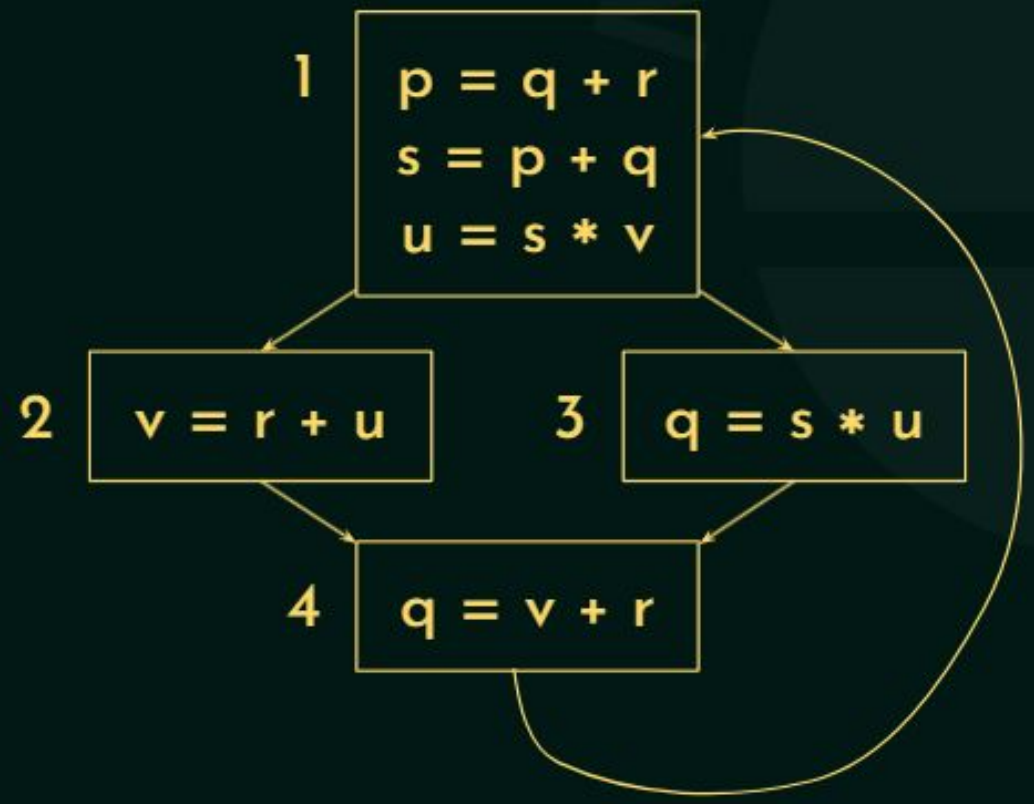
2. **Worklist Initialization:** Create a worklist containing all the Basic Blocks of the CFG.

3. **Iterative Dataflow Analysis:** While the worklist is not empty, perform the following steps for each basic block:
   - Calculate IN[B] = USE[B] ∪ (OUT[B] - DEF[B])
   - Calculate OUT[B] = ∪ IN[S]

4. **Final Liveness Information:** After the analysis has converged, the 'IN' sets represent the live variables at the entry points of each block.

Q: A variable x is said to be live at a statement S_i in a program if the following three conditions hold simultaneously:

1. There exists a statement S_j that uses x
2. There is a path from S_i to S_j in the flow graph corresponding to the program
3. The path has no intervening assignment to x including at S_i and S_j



The variables which are live both at the statement in basic block 2 and at the statement in basic block 3 of the above control flow graph are

a. p, s, u
b. r, s, u
c. r, u
d. q, v

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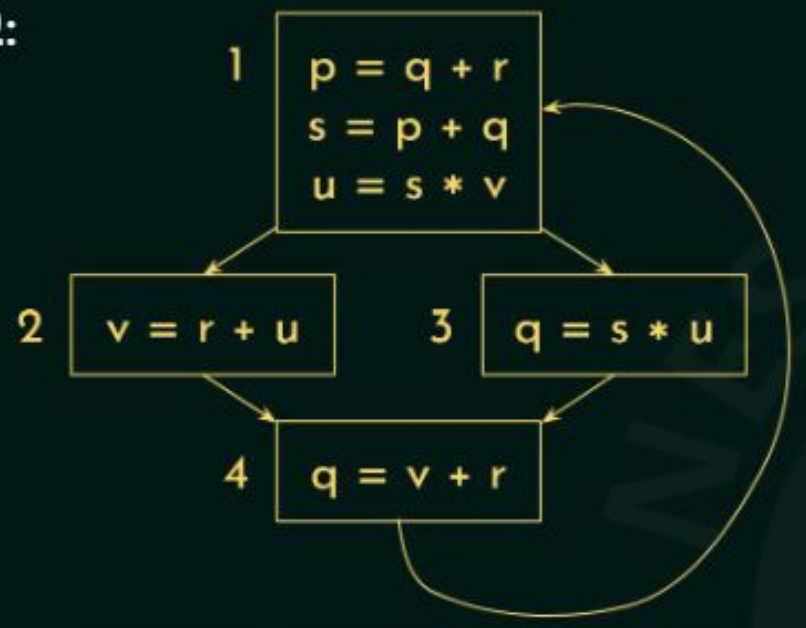
Q:



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   - Calculate IN[B] = USE[B] U (OUT[B] - DEF[B])
   - Calculate OUT[B] = U IN[S]
4. Final Liveness Information: After the analysis has converged, the 'IN' sets represent the live variables at the entry points of each block.

Sol.

Basic Block	USE	DEF	IN	OUT	IN	OUT	IN	OUT
1	q, r, v	p, s, u	q, r, v	r, u, s	r, q, v	r, u, s, v	r, v, q	r, u, s, v
2	r, u	v	r, u	v, r	r, u	r, v	r, u	r, v
3	s, u	q	s, u	v, r	v, r, s, u	r, v	r, v, s, u	r, v
4	v, r	q	v, r	q, r, v	r, v	r, q, v	r, v	r, v, q

Q: A variable x is said to be live at a statement S_i in a program if the following three conditions hold simultaneously:

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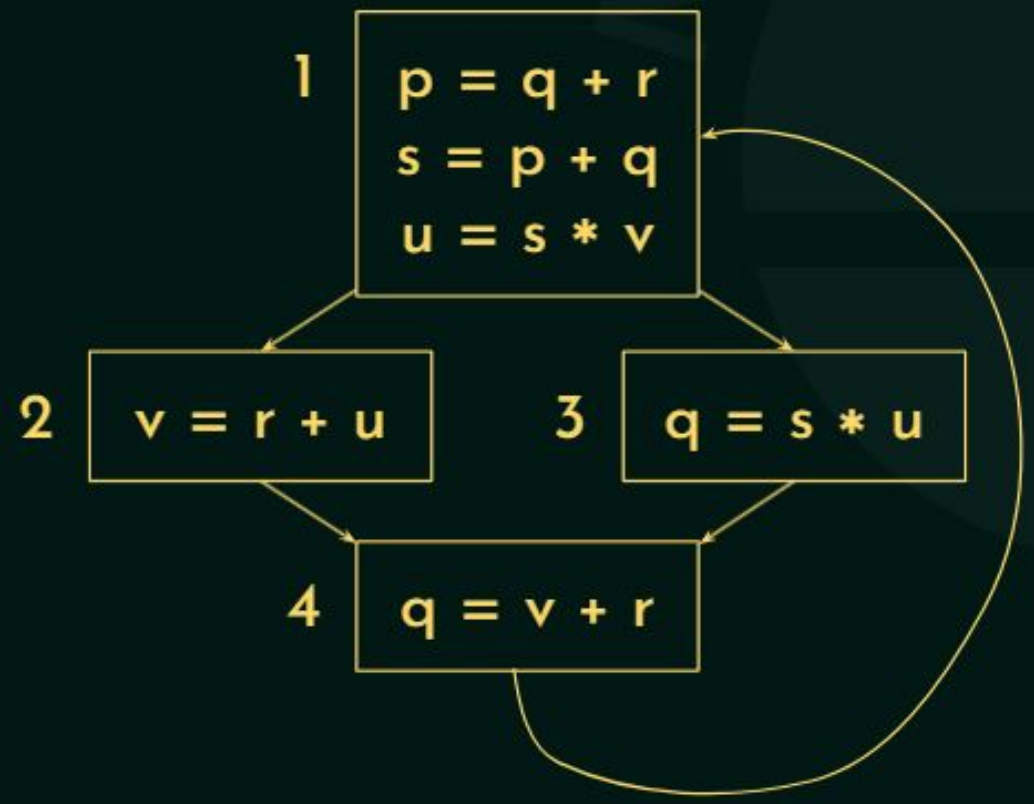

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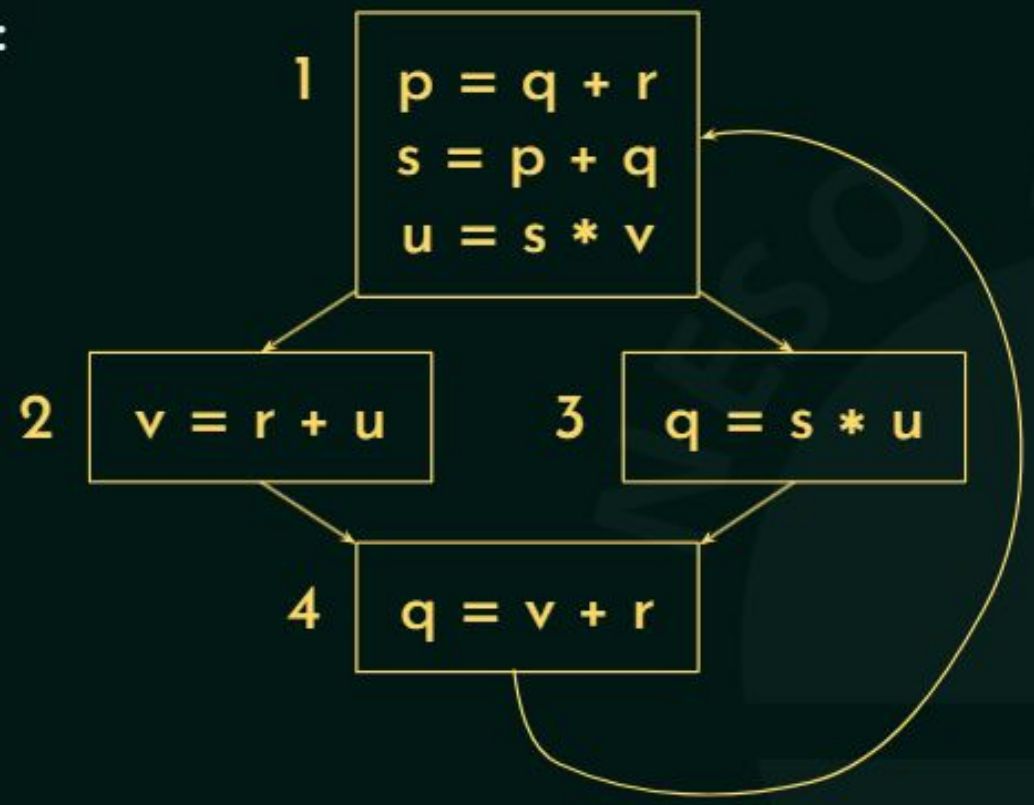

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c. r, u

d. q, v

Sol.

Basic Blocks	p	q	r	s	u	v
2	✗	✗	✓	✗	✓	✗
3	✗	✗	✓	✓	✓	✓